

SNEHAL LAL DAS

ML Engineer · GenAI Developer · MLOps Engineer

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PROFESSIONAL SUMMARY

Final-year CS (AI & ML) student with a track record of building and shipping production ML systems independently. Architected a 270M-parameter Transformer from scratch, deployed a live AI-powered security tool (CodeSentinel), and engineered an end-to-end MLOps pipeline with drift detection, CI/CD, and real-time monitoring. Seeking full-time ML Engineer / GenAI Developer roles where I can own the full model-to-production lifecycle.

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript

ML / DL: PyTorch, Scikit-learn, Transformers, CUDA, HuggingFace (familiar)

GenAI & LLM: RAG, Prompt Engineering, Embeddings, Vector Databases, ChromaDB, Mistral AI, Mistral Codestral, Semantic Chunking, LangChain (familiar), Fine-tuning (familiar)

MLOps & DevOps: Docker, Kubernetes, Jenkins, Apache Airflow, CI/CD, Git, Linux

Monitoring: Prometheus, Grafana, Alibi-Detect, Redis

Backend: FastAPI, Flask, REST APIs, JWT, SQLite, SQLAlchemy

Cloud & Deployment: Google Cloud Platform (GCP), Render, GitHub Pages

Data: Pandas, NumPy

PROJECTS

CodeSentinel — AI Code Vulnerability Scanner

Feb 2025 — Apr 2025

Stack: Python · FastAPI · Mistral Codestral · SQLite · JWT · GitHub Pages [GitHub ↗](#) | [Live Demo ↗](#)

- Architected and shipped a **production-grade AI security tool** that detects **4+ vulnerability classes** (SQL injection, hardcoded secrets, missing validation, insecure auth) across **8 programming languages** using Mistral Codestral.
- Reduced manual code review effort by automating severity triage — classifying findings into **Critical / High / Medium / Low** with concrete, per-finding fix suggestions, cutting remediation lookup time significantly.
- Engineered a secure backend with **JWT authentication**, bcrypt password hashing, and a full scan-history REST API; achieved zero reported auth vulnerabilities post-deployment.
- Owned the full product lifecycle — frontend (GitHub Pages), backend (Render), and CI pipeline — delivering a publicly accessible tool solo in **under 3 months**.

Codebase RAG System

Jan 2025 — Mar 2025

Stack: Python · ChromaDB · Mistral AI · sentence-transformers · FastAPI [GitHub ↗](#)

- Designed and deployed a **production-grade RAG pipeline** enabling natural-language querying across large multi-file codebases, reducing codebase navigation time by an estimated **~60%** vs. manual search.
- Increased retrieval precision by **~40%** over naive line-based splitting by implementing language-aware semantic chunking via **AST parsing** (Python) and regex (JS/TS).
- Optimised vector search with ChromaDB and sentence-transformer embeddings to achieve **sub-2s end-to-end query latency**; integrated Mistral AI for contextual answers with file-path and line-level citations.

MLOps-Enabled ML Pipeline — Survival Prediction

Aug 2024 — Nov 2024

Stack: Python · Scikit-learn · Airflow · Redis · Docker · Prometheus · Grafana [GitHub ↗](#)

- Delivered an end-to-end MLOps pipeline with a Random Forest classifier achieving **86% accuracy** on held-out test data, with **11 engineered features** improving baseline performance by ~12%.
- Eliminated **~35% of redundant inference computation** by architecting a Redis-based feature store; automated retraining and feature engineering workflows using Apache Airflow DAGs.
- Reduced model degradation risk by integrating **Alibi-Detect** for real-time data drift detection, with automated retraining triggers surfaced via Prometheus + Grafana dashboards.

- Containerised a Flask REST API (/predict, /metrics) with **Docker** for fully reproducible, environment-agnostic deployment across dev and production.

Custom Transformer Language Model — 270M Parameters

Apr 2024 – Jun 2024

Stack: PyTorch • Python • CUDA • T4 GPU [GitHub ↗](#)

- Architected and trained a **270M-parameter Transformer LLM from scratch** in PyTorch implementing Gemma-style architecture: GQA, RoPE positional embeddings, RMSNorm, and gated FFN.
- Maximised GPU throughput to **~100% T4 utilisation** (~9 GB VRAM) via gradient accumulation and mixed-precision (AMP) training, sustaining **~8,000 tokens/sec** across 29,000 training steps.
- Achieved **validation loss of 2.3083**; built a resume-capable checkpointing system and deployed the trained model via a FastAPI inference API with CPU-compatible serving.

EDUCATION

BTech — Computer Science Engineering (AI & ML)

Sep 2022 – Jun 2026 (Expected)

Roorkee Institute of Technology, Uttarakhand, India

CGPA: 6.68 / 10 (through Sem 6) · **Relevant coursework:** Machine Learning, Deep Learning, Data Structures & Algorithms, Database Systems, Cloud Computing

CERTIFICATIONS

- TensorFlow for Deep Learning Bootcamp — Udemy
- MLOps on GCP: CI/CD, Kubernetes, Jenkins — Udemy
- Introducing Generative AI with AWS — Udacity